Fundamentals of Accounting

FINANCIAL MANAGEMENT

OBJECTIVES

Understand:

- The basic concepts of accounting and costing
- Why there is the need for financial accounting
- The various formats of summarising accounting information, and information reported on financial statement.
- Financial Ratio Analysis



ACCOUNTING

- Accounting is the process of **identifying, measuring, recording and communicating** financial information about an organization's activities.
- An **accounting system** is that which is used to manage the accounting process.
  - Usually computer systems and software packages are used
  - E.g. NetSuite – dedicated accounting system
    - Enterprise resource planning (ERP) software packages like SAP which have as part of them accounting system

Bookkeeping

- A component of accounting.
- It is the process of recording information and maintaining accounting records.
- Accountants are the key players in the design of accounting systems used in businesses.
- They also design reports to meet various needs for financial information-
  - Financial statements
  - Tax returns
  - Budgets and
  - Special cost analysis



DEMAND FOR ACCOUNTING INFORMATION

Accounting information is useful because it helps people answer questions and make better decisions.

Accounting information may be demanded by:

- People who are within the enterprise (who are the key people? and why would they demand accounting information?).
- People who are external to the enterprise (who are they and why would they want accounting information?).

INSIDE DEMAND

- The principal demand for accounting information from inside the organization comes from :-
- MANAGERS need accounting information to control both human and material resources and to direct resources to the most effective uses.
- Managers use accounting information to predict the consequences of their future actions and to help decide which actions to take
- Managers use accounting to evaluate the performance of employees
- INVESTORS use accounting information to decide where to invest their money
- EMPLOYEES use accounting information about their organization to help them plan their careers.

OUTSIDE DEMAND

The principal demand for accounting information from outside the organization comes from :-

- CAPITAL MARKETS help arrange the investment of billions of dollars each year. Bank loan officers use it as a basis for lending decisions
- GOVERNMENTS need accounting information to determine taxes owed businesses, to implement a variety of regulatory objectives and to formulate economic policy.
- SPECIAL INTERST GROUPS including consumer action groups, labour unions, customers, competing businesses, financial advisers, and the general public – seek accounting information as a basis for furthering their own interests.

Meeting the demand for accounting information

- The demand for accounting information, whether it comes from outside or inside an organization is met by the same accounting system.
- The data are summarised and reported in different ways to meet different needs for information.
- Accounting and reporting to satisfy the inside demand for accounting information is called **management accounting**.
- Accounting and reporting to satisfy the outside demand for accounting information is called **financial accounting**.

QUESTIONS ANSWERED BY FINANCIAL STATEMENTS

Financial statements summarise the performance and status of a business.

- They are issued at least annually.
- Some of the questions that are answered by the financial statements prepared for outsiders are as follows:
  - How much better off is the business at the end of the year than it was at the beginning?
  - What are the economic resources of the business and the claims against those resources?
  - How much cash flowed in and out of the business during the year?

Reporting business activities on financial statements

- The typical business entity issues four financial statements.
- They are:
  - **Income statements/profit and loss statement**
  - **Balance sheet**: a statement of the assets against liabilities over a specified period of time
  - **The statement of cash flows**
  - **Statement of changes in retained earnings/Owners Equity**

Income statement

- **Income statements** – the purpose of income statement is to report to internal and external users the revenue earned and expenses incurred during a particular time period and the net income (or loss) generated by these events.
- It is sometimes called profit and loss statement or a summary of operations
- Income statements can be used to identify certain trends when analysed over several periods of time.
- Income statement is a period of time description of a business and they may be prepared quarterly or monthly as well as annually

INCOME STATEMENT: AN EXAMPLE

Oguaa Retail Stores Co Ltd
Income Statement
For the year ended July, 31, 2012

Net Sales	1200, 000
Cost of goods sold	850, 000
Gross profit	350, 000
Selling, general, and administrative expenses	311, 000
Earnings from operations	39, 000
Interest Expense	9000
Earnings before tax	30000
Income tax	12000
Net Income	18000

Balance sheets

- A balance sheet is a statement that shows the **financial position** of a business entity **at a point-in-time** (a snapshot).
- It is a statement of the total assets and liabilities of an organization at a particular date.

The whole of financial accounting is based on the **ACCOUNTING EQUATION**. This equation is based on the fact that for a firm to operate, it needs resources. These resources are known as **ASSETS** and they need to be supplied by **CAPITAL** provided by the business owner(s). The following equation would hold:

ASSETS = CAPITAL

ACCOUNTING EQUATION

- If some of the assets are provided by others other than the owners, the indebtedness of the firm for these resources is known as **liabilities.**
- The accounting equation then becomes:

Assets = Capital + Liabilities

- The totals of each side will always be equal to one another
- Assets consist of property of all kinds such as buildings, machinery, stocks of goods and bank accounts.
- Liabilities consist of amounts owed for goods supplied to the firm and loans made to the firm.
- **Capital** is generally called **owners equity** or **net worth**.



LONG TERM AND CURRENT

- To provide additional information to the user, assets and liabilities are usually classified in the balance sheet as:
  - **CURRENT ASSETS**:- those due to be repaid or converted into cash within 12 months of the balance sheet date.
  - **LONG-TERM**:- those due to be repaid or converted into cash more than 12 months of the balance sheet date.
- **FIXED ASSETS**:- these are assets which are of a permanent nature and is used by the business to provide the capability to conduct its trade.
  - Examples are plant and machinery, land and buildings, motor vehicles, patents, trademarks and brands.



CAPITAL

- **CAPITAL OR EQUITY CAPITAL** is the financing that owners of a business put in to finance their business.

- Undistributed profits are reinvested in company assets, eg. Stocks, equipment, and bank balance and added to equity capital of the business.

- Retained earnings may also be paid out as dividends to shareholders at a future date.

- At any time , the capital of a business is equal to the assets received from the owners plus any profit made by the company that remains undistributed.

Balance sheet: an example

Introduction of capital

- On Jan 1, 2009 K. Ananse started a business and deposited GH₵10,000 in his bank account opened specially for the business. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 1 January 2009

Assets	GH₵	**Capital and Liabilities**	GH₵
Cash at Bank	10,000	Capital	10,000
	10,000		10,000

THE PURCHASE OF AN ASSET BY CHEQUE

- On Jan 3, 2009 K. Ananse buys a building for GH₵4,000 and pays for it with a cheque. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 3 January 2009

Assets	GH₵	**Capital and Liabilities**	GH₵
Building	4,000	Capital	10,000
Cash at Bank	6,000		
	10,000		10,000

THE PURCHASE OF AN ASSET AND THE INCURRING OF A LIABILITY

- On Jan 6, 2009 K. Ananse buys some goods for GH₵1,000 from K Kuaso and agrees to pays for them sometime within the next two weeks. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 6 January 2009

Assets	GH₵	Capital and Liabilties	GH₵
Building	4,000	Capital	10,000
Stock of goods	1000	Liabilities	
Cash at Bank	6,000	Creditor – K Kuaso	1,000
	11,000		11,000

THE SALE OF AN ASSET ON CREDIT

- On **Jan 10, 2009**, goods which cost GH₵ 500 were sold to A. Ntikuma for the same amount of money to be paid later. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 10 January 2009

Assets	GH₵	Capital and Liabilties	GH₵
Building	4,000	Capital	10,000
Stock of goods	500		
Debtor	500	Liabilities	
Cash at Bank	6,000	Creditor – K Kuaso	1,000
	11,000		11,000

THE SALE OF AN ASSET FOR IMMEDIATE PAYMENT

- On Jan 13, 2009, goods which cost GH₵ 200 were sold to Y. Okondor for which he paid for immediately by cheque. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 13 January 2009

Fixed Assets	GH₵	Capital and Liabilties	GH₵
Building	4,000	Capital	10,000
Current Assets			
Stock of goods	300		
Debtor	500	Liabilities	
Cash at Bank	6,200	Creditor – K Kuaso	1,000
	11,000		11,000

Balance sheet- payment of a liability

- On Jan 15, 2009, K Ananse pays a cheque of GH₵ 500 to K Kuaso in part payment of the amount owed. The balance sheet would appear:

K Ananse Co. Ltd.

Balance Sheet as at 15 January 2009

Fixed Assets	GH₵	Capital and Liabilties	GH₵
Building	4,000	Capital	10,000
Current Assets			
Stock of goods	300		
Debtor	500	Liabilities	
Cash at Bank	5,700	Creditor – K Kuaso	500
	10,500		10,500

BALANCE SHEET- COLLECTION OF AN ASSET

A NTIKUMAH WHO OWED K ANANSE GH₵ 500 MAKES A PART PAYMENT OF GH₵200 BY CHEQUE TO K ANANSE ON 31 JAN. 2009 . THE BALANCE SHEET LOOKS LIKE THIS:

K Ananse Co. Ltd.

Balance Sheet as at 31 Jan. 2009

Fixed Assets	GH₵	**Capital and Liabilities**	GH₵
Buildings	4,000	Capital	10,000
Current assets			
Stock of Goods	300		
Debtor	300	**Liabilities**	
Cash at Bank	5,900	Creditor – K Kuafo	500
	10,500		10,500

Effects of transactions on the accounting equation

- It can be seen that every transaction affects two items on the accounting equation.
- Sometimes it has changed two assets by reducing one and increasing the other.
- More times than not, it affects both sides of the equation.



Effects of transactions on the accounting equation

Example of Transaction	Effects on Particular Assets	Effects on Liability/Capital
Buy goods on credit		
Buy goods by cheque		
Pay creditors by cheque		
Owners pay more money into the bank		
Owner takes money out of the business bank for his own use		
Owner pays creditor from private money outside the firm		

Financial ratio analysis

- Financial ratio analysis is an analysis of financial statements conducted by preparing and evaluating a series of ratios.
- Ratios provide meaningful information only when compared with ratios for the same firm using previous statements or similar firms.
- Ratios help analysts by removing most of the effects of size differences between firms.
- Financial ratios can be divided into four basic types:
  - Liquidity
  - Debt
  - Profitability
  - Activity



LIQUIDITY RATIO

- used to judge how well an organization will be able to meet its short term financial obligations.
- Short term creditors want to know the likelihood that a business will be able to pay its obligations as they come due.
- The cash to pay liabilities as they mature will come from the business's available cash and from assets that can be readily converted into cash. Receivables, inventories, marketable securities can all be converted into cash relatively quickly.
- Property, plant and equipment and other long-term assets are liquidated much more slowly.

Liquidity ratios

- Two ratios typically used are:
  - Current ratio
  - Quick ratio or acid-test ratio
- Both focus on the relationship between current assets and current liabilities.
- The quick ratio is a more conservative (cautious) measure of liquidity because it excludes inventory from the calculation.



Current Ratio

- Used to analyse the company's working capital conditions
- Expressed as an equation:
  - **Current ratio = $\frac{\text{Total current assets}}{\text{Total current liabilities}}$**
- Since current liabilities are those debts payable in the next year, a current ratio value of 3 means that current assets will cover shot-term debts three times. Current ratio values of 2-3 are common and desirable.
- For comparison, it is necessary to compute the current ratio and all other ratios for several companies in the same industry



Acid – Test Ratio (quick ratio)

- Some analysts believe that the current ratio overstates short-term liquidity (ie. working capital invested in inventory can be converted into cash rapidly).
- They argue that prepaid expenses (expenses for which payments are made before consumption) often cannot be converted into cash. Further, inventories must be sold and receivables collected from those sales before cash is obtained to pay current liabilities.
- Both sale of inventory and the collection of receivables take time.
- Conservative analysts argue that only those current assets that can be turned into cash almost immediately should be used to measure short-term liquidity.

Acid – Test Ratio (quick ratio)

- Mathematically:

- Acid-test Ratio = $\frac{\text{Current Assets – Inventories}}{\text{Current liabilities}}$

  - An acid test ratio of **1.0** is considered sound and this is because there are just enough liquid assets available to satisfy current liabilities.

  - If an organisation cannot meet its short-term debt, it often has to go to a high-cost lender for financial assistance.

Debt Ratio or Leverage Ratio

- It measures the proportion of owners and creditors claim on the organization relative to their assets
- It indicates the organization's ability to meet long term obligations
- Common debt ratios
  - The debt-to- equity ratio and
  - The total debt to total asset ratio
- These ratios will indicate the proportion of debt in the capital structure
- Debt to equity ratio = $\frac{\text{Total Liabilities}}{\text{Owners equity}}$
- Debt Ratio = $\frac{\text{Total Liabilities}}{\text{Total Assets}}$

The higher the ratio, the more risky the business.

Profitability Ratio

- Measures the performance of a firm
- Two typical ratios used are"
  - Return on Investments (ROI)/ Return on Assets
  - Return on Equity

- Return on Investment = $\frac{\text{Net Income}}{\text{Average assets}}$
- Return on Equity = $\frac{\text{Net Income}}{\text{Average owners equity}}$
- Earnings Per Share: $\frac{\text{Net Income}}{\text{Number of common shares outstanding}}$
- Note: We are using average assets and average owners equity
- Result of the analysis is presented in a percentage format

Activity (or Operating)Ratio

- An activity ratio assesses how effectively an organisation can generate revenue from its balance sheet accounts.
- Examples include:
  - Inventory turnover ratio: Provides an indication of whether a company has excessive or inadequate stock of goods or inventory
  - **Inventory turnover ratio** = Cost of goods sold / Average inventory

  Divide 365 by whatever value you get
  - Account receivable turnover: indicates the average length of time (in days) a business must wait before it receives payment for good sold.
  - **Account receivable ratio** = Net receivable / Average receivables
    - The total money owed to a company by its customers, minus the money owed that will likely never be paid
    - Number of days in receivables = 365 / Turnover ration

CRUISERS, INC.

Comparative Condensed Balance Sheets

September 30, 1996 and 1995

	1996	***1995***
Current assets:		
Cash and marketable securities	$22,286	$16,996
Account receivable	42317	39620
Inventories	53716	48201
Total current assets	$118,319	$104,817
Other assets	284335	259903
Total assets	$402,654	$364,720
Current liabilities	$57,424	$51,400
Other liabilities	$80,000	$83,000
Total liabilities	$137,424	$134,400
Owner's equity	$265,230	$230,320
Total liabilities and owner's equity	$402,654	$364,720

CRUISERS, INC

Condensed Income Statement

For the Year Ended

30-Sep-96

Net sales	$611,873
Cost of goods sold	28354
Gross margin	183519
Operating expenses	122183
Earnings before interest and taxes	$61,336
Interest expenses	6400
Earnings before taxes	54936
Income tax	22000
Net income	32936
Earnings per share	$1.21

Using the Balance sheet and income statement of Cruisers, Inc provided above, calculate the following financial ratio:

- Liquidity ratios
    1. Current ratio
    2. Acid test ratio
- Debt ratio
    1. Debt ratio
    2. Debt/equity ratio
- Profitability ratios
    1. Return on investment
    2. Return on equity
- Activity measures
    1. Account receivables ratio
    2. Inventory turnover



Cost Accounting (CA)

It is the process of accumulating the information managers need to identify, measure and control costs.

-It involves accounting for the cost of products and services

-It is useful for evaluating managerial performance

Objectives of Cost Accounting

-To assist in pricing the product or service.

-To assist in the measurement of reported profits

-To assist in planning decisions-which product to manufacture, their quantities etc

Cost Accounting: Basic Terminologies

Direct Labour: Costs of salaries, wages and benefits for staff who work direct on manufacture products.

Indirect Labour: Cost of personnel who do not work directly on the product but whose services are necessary for the manufacturing process.

Sunk Cost: Retrospective costs that cannot be recovered.



Allocation of Factory Expense (Overhead)

All costs incurred in the production of an item are accounted for by the cost accounting system.

The cost accounting system considers the following costs:

- Material Cost
- Direct Labour Cost
- Factory Expense Cost (Overhead)

Factory Cost = Material Cost + Labour Cost + Factory Expense Cost

- Allocation schemes are used to distribute the expenses on a reasonable basis.

Factory Expense Allocation Basis

Overhead Category	Allocation Basis
Taxes	Space Occupied
Utilities	Space, direct labour hours, direct labour cost
Salaries	Direct labour hours, direct labour cost

The most common basis is to use direct labour cost and direct labour hours



Factory Expense Allocation Basis

Most allocation is done using a predetermined factory expense rate:

$$= \frac{\text{Estimated overhead cost}}{\text{Estimated basis level}}$$

Example: Factory Expense Allocation

The following information table was obtained from last year's budget for the three machines used to produce bookends. Determine the overhead rate for each machine if the estimated factory expense budge for producing the bookends is $ 5000 per machine.

Cost Source	Allocation Basis	Estimated Activity Level
Machine 1	Labour cost	$ 10, 000
Machine 2	Labour hours	2, 000 hours
Machine 3	Material Cost	$ 12, 000

Solution: factory expense allocation problem

Over head rate = Estimated overhead cost / Estimated basis level

Overhead Rate for Machine 1 = $ 5000 / $ 10, 000 = $ 0.5 per labour

Overhead Rate for Machine 2 = $ 5, 000 / 2,000 = $ 2.5

Overhead Rate for Machine 3 = $ 5, 000 / $12, 000 = $ 0.42 per unit of material

Determination of variance

After determining the overhead rate, it is possible to determine the cost of production.

-This may be for a product or service, or the cost of operating a department.

-If the budget is correct, then the estimated overhead = allocated overhead

-However, because of errors in budgeting, there is usually a variance.

-i.e. the difference between the allocated overhead and the estimated overhead.



Variance Calculation: An example

Determine the total factory overhead expense , factory cost and the variance using the following actual data given in the table below:

Cost Source	Machine Number	Actual Cost ($)	Actual Hours
Material	1	3800	
	3	19,550	
Labour	1	2, 500	650
	2	3,200	750
	3	2, 800	720

Note: Consult first example for overhead rates



Solution

Factory cost = material cost + labour cost + factory expense (overhead)

Machine 1 overhead = (labour cost)(rate) = ($2,500)(0.5)
= $1,250

Machine 2 overhead = (labour hours) (rate) = ($750h) ($2.50/h)
= $1,875

Machine 3 overhead = (material cost) rate = ($19,550 ((0.42)
= $8,211

Thus total factory expense = $ (1,250 + 1875 + 8211) = $ 11, 336

Factory cost = $(3,800 + 19,550 + 2,500 + 3,200 + 2, 800 + 11, 336)
= $43, 186

Variance = (Estimated overhead) – (Actual overhead)
= 3 * ($ 5000) – $ 11, 336
= $ 3,664

The budget of $15000 overhead represents a 32.3% over the actual overhead